

Mikhail Razakov

Static Analysis / Program Analysis Engineer

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Static-analysis engineer with professional C#.NET context at ISP RAS and LLVM 22 experience from MCST. DerefAfterNullAnalyzer implements intraprocedural fixed-point data flow over Roslyn CFG with branch refinement, predecessor joins, and loop/back-edge iteration; Global-IV adds interprocedural effect analysis and affine evolution.

EXPERIENCE

- 2026 — present** **ISP RAS — Static Analysis Engineer**
- Contribute to SharpChecker, ISP RAS's C#.NET static-analysis platform.
- July — August 2026** **MCST — Compiler Engineering Intern**
- Implemented a LICM pass for LLVM 22 in C++; hoisting conditions check loop structure, memory access, side effects, and speculative safety.
 - Built a Python harness comparing program behavior before/after optimization and checking transform/no-transform cases.

PROJECTS

DerefAfterNullAnalyzer

A Roslyn ControlFlowGraph analyzer propagates null state on conditional edges to a fixed point, joins predecessor states, and reprocesses successors for loops/back edges.

Method/property/field/index/event dereferences; invalidation after assignment/ref/out; dedicated analyzer tests use Microsoft.CodeAnalysis.CSharp.Testing.

LLVM Interprocedural Global IV Optimization

An LLVM 22 pass localizes supported induction-like globals after interprocedural effect analysis, affine evolution, and a separate applicability check; APInt models fixed-width wraparound.

29 positive/negative regression cases cover LLVM Verifier checks, structural expectations, idempotence, and observable before/after behavior.

SKILLS

Data-flow Analysis

Roslyn CFG, worklist/fixed point, edge refinement, predecessor joins, loops/back edges

Interprocedural / LLVM

effects analysis, affine evolution, APInt, transformation applicability, LLVM IR

Analysis Correctness

assignment/ref/out invalidation, conservative unsupported-case handling, regression tests, LLVM Verifier

Languages / Tools

C#, C++23, Roslyn, LLVM 22, .NET

EDUCATION

HSE University — Software Engineering, 2026–2030

RECOGNITION

LangDev'26 — Build the Language, Then Make the Abstractions Disappear

Accepted talk on UniversalToolchain's architecture; LangDev'26 takes place October 8–9, 2026 in Málaga.

UniversalToolchain — Baltic Science and Engineering Competition

First Degree Diploma and Grand Prize "Perfection as Hope" for UniversalToolchain.

MEPhI Junior

Overall winner; First Degree Diploma, 96/100 for UniversalToolchain.

Moscow / remote